Calibration results
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Normalized Residuals

Reprojection error (cam0): mean 0.2674842236470971, median 0.22450173763913364, std: 0.18299366061916697
Reprojection error (cam1): mean 0.24693788662943664, median 0.21131385988665302, std: 0.16124077382385407
Gyroscope error (imu0): mean 1.4601050501632562, median 1.3149692259829573, std: 0.8977354826446917
Accelerometer error (imu0): mean 2.9430969831759266, median 2.359770661988833, std: 2.6847551639459306

Residuals

Reprojection error (cam0) [px]: mean 0.2674842236470971, median 0.22450173763913364, std: 0.18299366061916697
Reprojection error (cam1) [px]: mean 0.24693788662943664, median 0.21131385988665302, std: 0.16124077382385407
Gyroscope error (imu0) [rad/s]: mean 0.0037851629111241803, median 0.0034089141345712303, std: 0.0023272812134489283
Accelerometer error (imu0) [m/s^2]: mean 0.038740884672874454, median 0.031062382107398354, std: 0.035340252385803274

Transformation (cam0):

T_ci: (imu0 to cam0):
[[0.99994159 -0.00261012 -0.01048813 0.006347]
[0.00262427 0.99999567 0.00133523 0.00738166]
[0.0104846 -0.00136268 0.99994411 -0.00756397]
[0. 0. 0. 1.]]

T_ic: (cam0 to imu0):
[[0.99994159 0.00262427 0.0104846 -0.0062867]
[-0.00261012 0.99999567 -0.00136268 -0.00737537]
[-0.01048813 0.00133523 0.99994411 0.00762026]
[0. 0. 0. 1.]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.002567402292028916

Transformation (cam1):

T_ci: (imu0 to cam1):
[[0.99999059 -0.00244572 -0.00358391 -0.04385591]
[0.00245118 0.99999584 0.00151949 0.00733702]
[0.00358018 -0.00152826 0.99999242 -0.00786446]
[0.   0.   0.   1.   ]]

T_ic: (cam1 to imu0):
[[0.99999059 0.00245118 0.00358018 0.04386567]
[-0.00244572 0.99999584 -0.00152826 -0.00745627]
[-0.00358391 0.00151949 0.99999242 0.00769608]
[0.   0.   0.   1.   ]]

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
-0.002576958503562673

Baselines:

Baseline (cam0 to cam1):
[[0.99997615 0.00017375 0.00690412 -0.05015182]
[-0.00017501 0.99999997 0.00018244 -0.00004215]
[-0.00690409 -0.00018364 0.99997615 -0.00025549]
[0.   0.   0.   1.   ]]
baseline norm: 0.05015249139192716 [m]

Gravity vector in target coords: [m/s^2]
[9.80638844 0.04720208 0.03066963]

Calibration configuration
=========================

cam0

Camera model: pinhole
Focal length: [400.0209447384781, 401.8797270027262]
Principal point: [325.1041382604159, 243.1256621683816]
Distortion model: radtan
Distortion coefficients: [0.01845057906179373, -0.010331358099150664, -0.0013031033977454912, 0.0019627603781516624]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.025 [m]
Spacing 0.0075 [m]

cam1

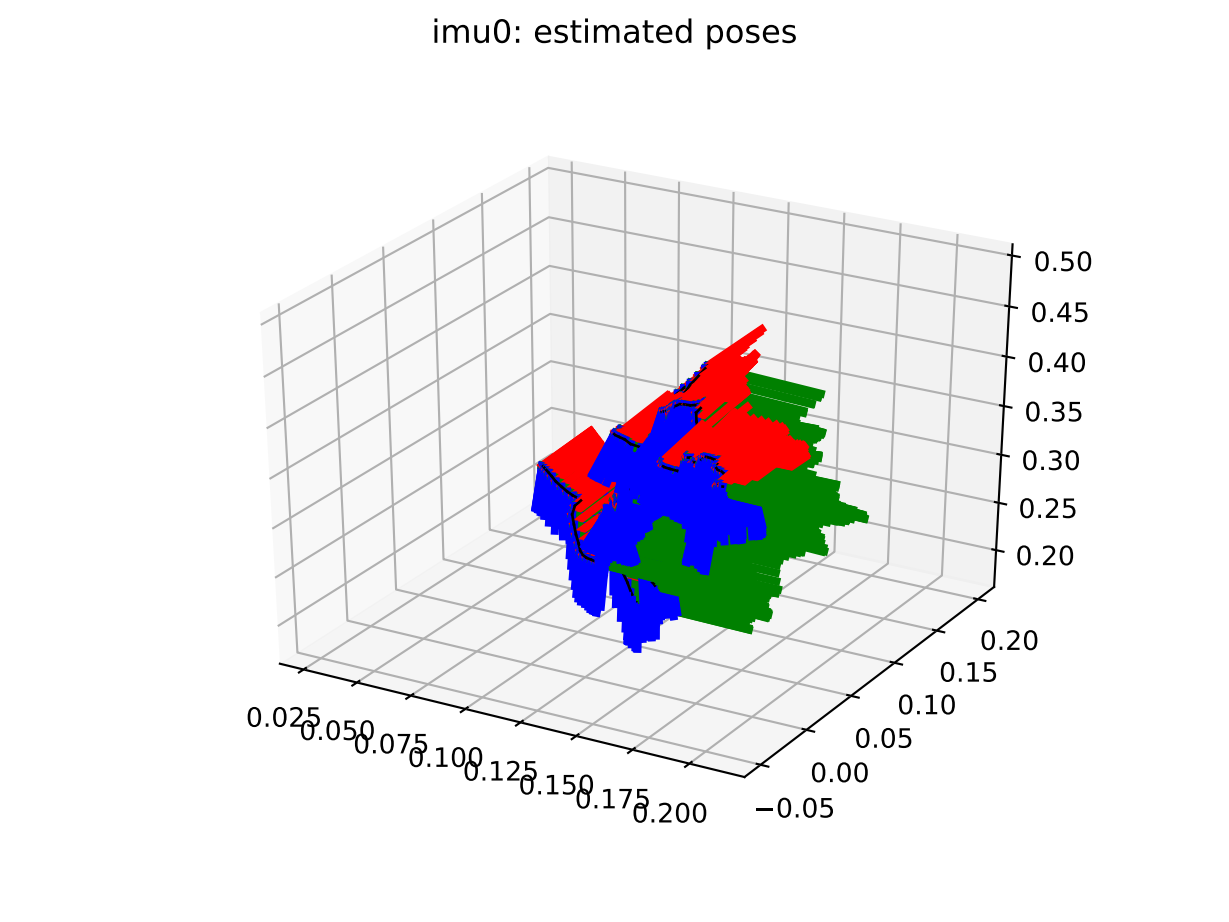
Camera model: pinhole
Focal length: [399.75858715252537, 401.6825004568713]
Principal point: [322.1724117647744, 243.21253352207617]
Distortion model: radtan
Distortion coefficients: [0.016959168374417948, -0.00782761582952835, -0.0014252593783064648, 0.0001746232658439117]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.025 [m]
Spacing 0.0075 [m]

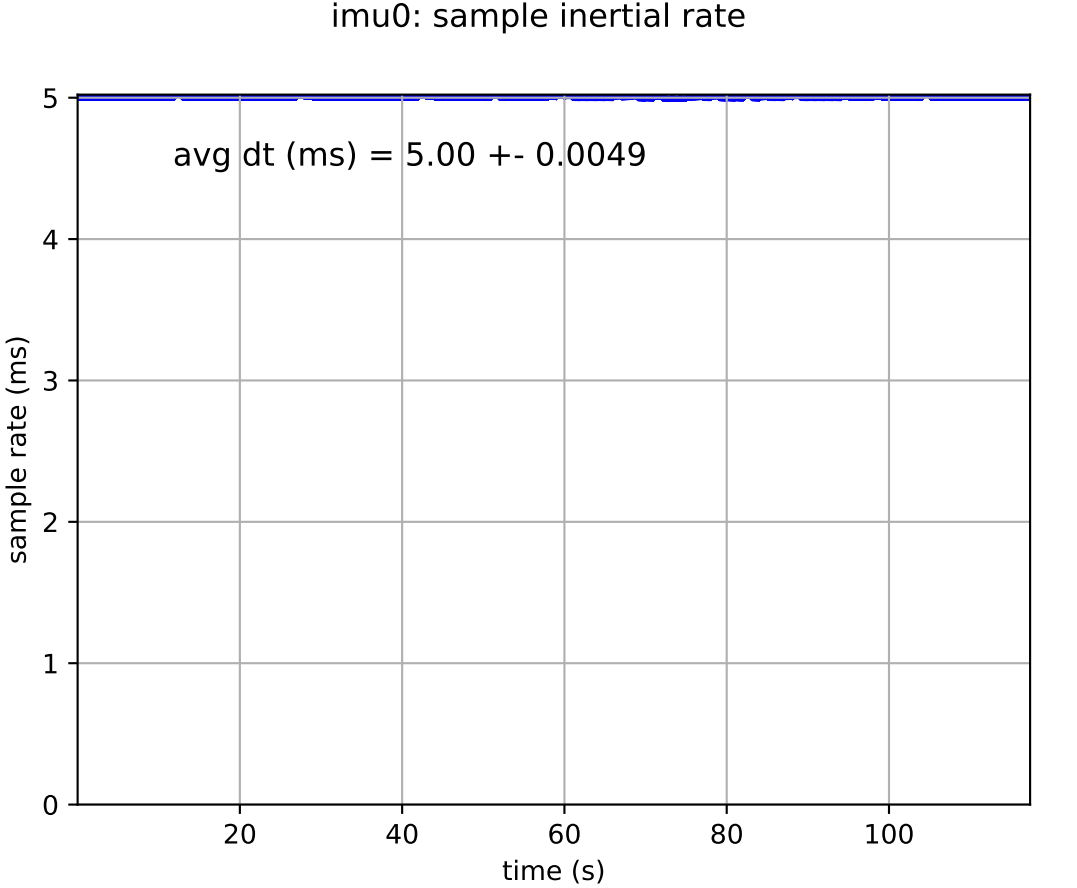
IMU configuration
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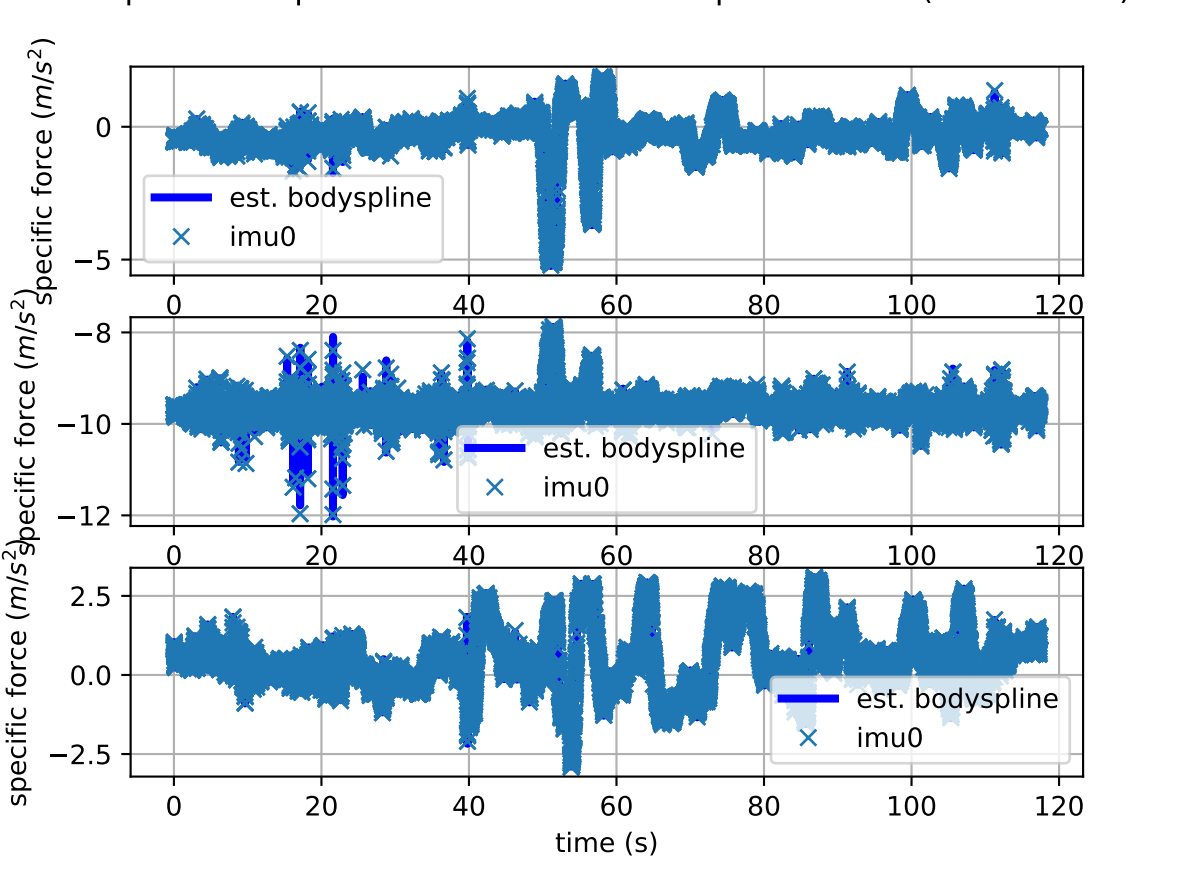
IMU0:

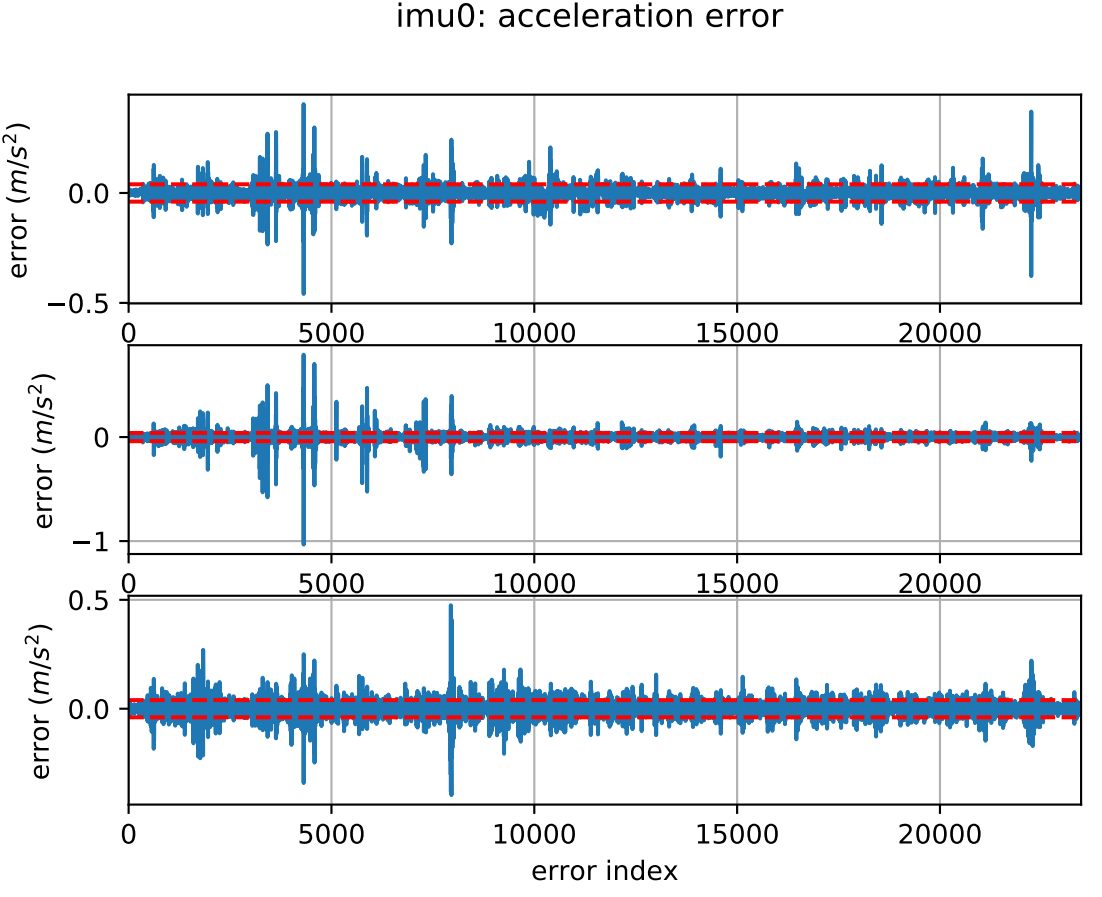
Model: calibrated
Update rate: 200.0

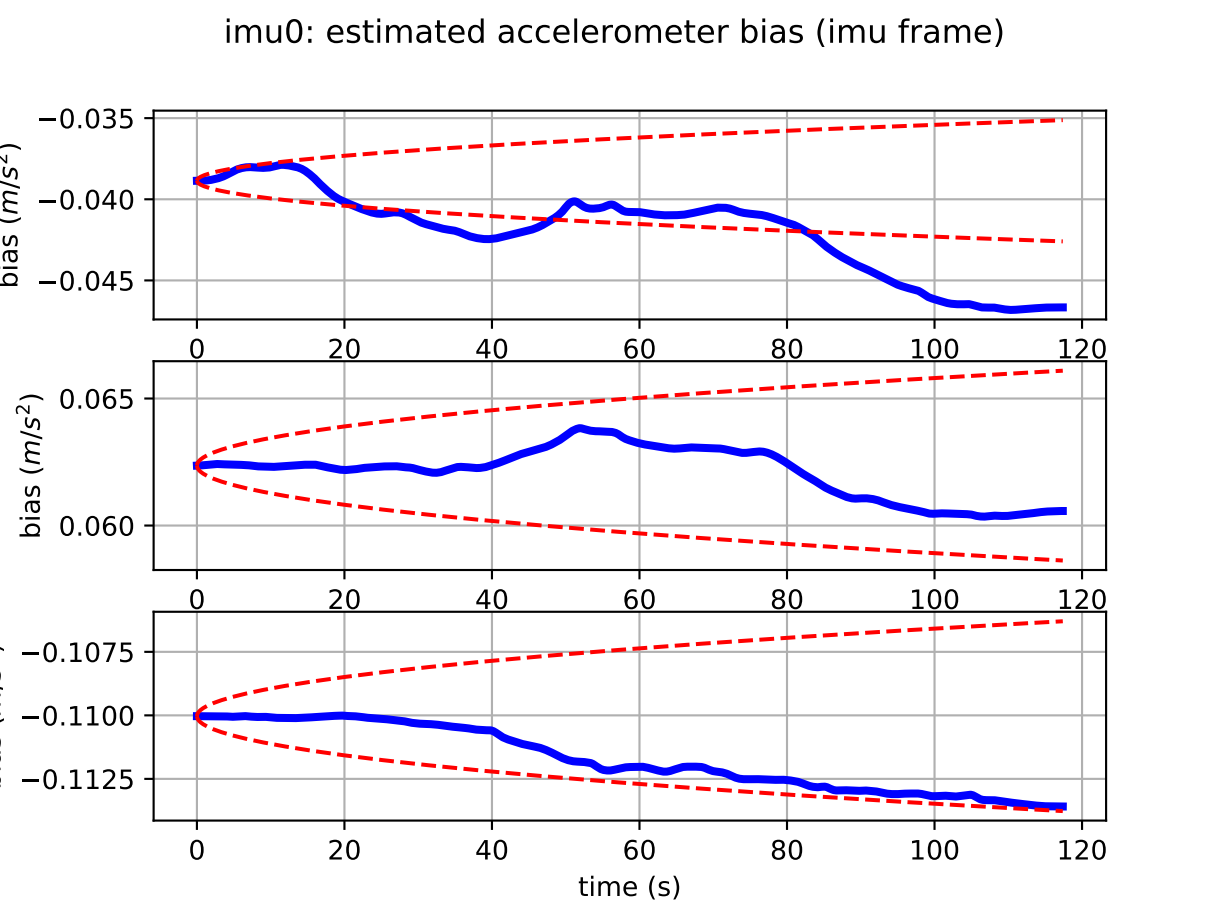
Accelerometer:
Noise density: 0.0009307862574
Noise density (discrete): 0.013163305488855746
Random walk: 0.0001149383323
Gyroscope:
Noise density: 0.000183309712
Noise density (discrete): 0.0025923908082510607
Random walk: 5.116486261e-06
T_ib (imu0 to imu0)
[[1. 0. 0. 0.]
[0. 1. 0. 0.]
[0. 0. 1. 0.]
[0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]

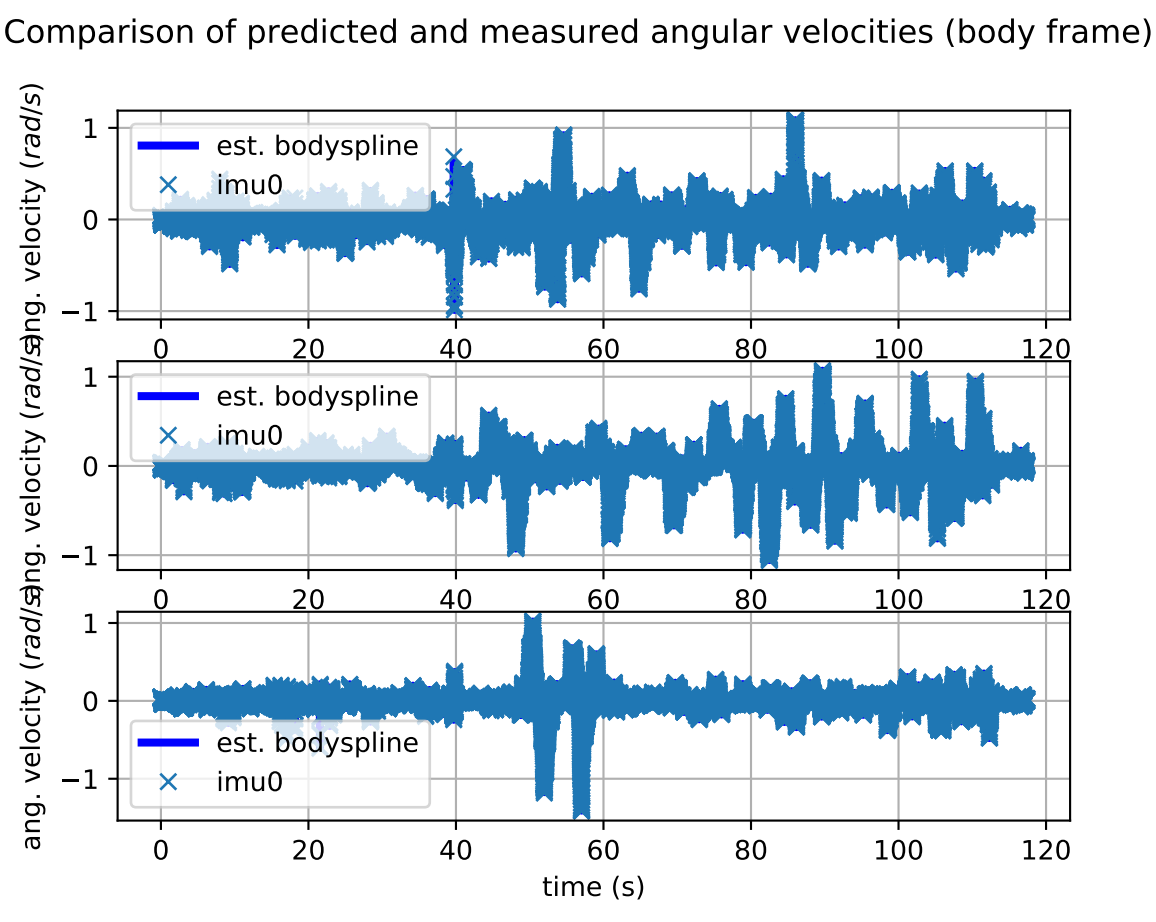

imu0: estimated poses
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0.20
0.15
0.10
0.05
0.00
−0.05
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200

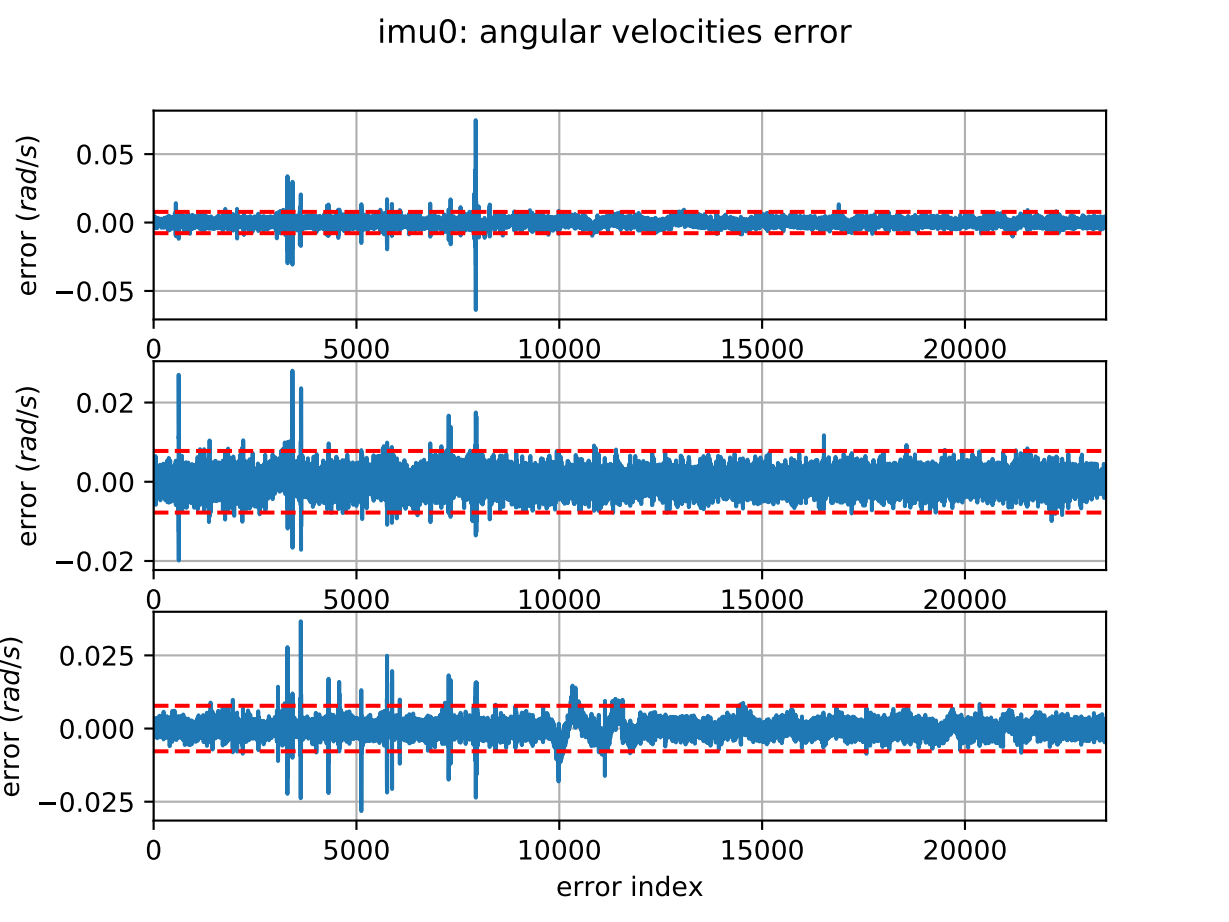

imu0: sample inertial rate
avg dt (ms) = 5.00 +- 0.0049
sample rate (ms)
time (s)
0
1
2
3
4
5
20
40
60
80
100

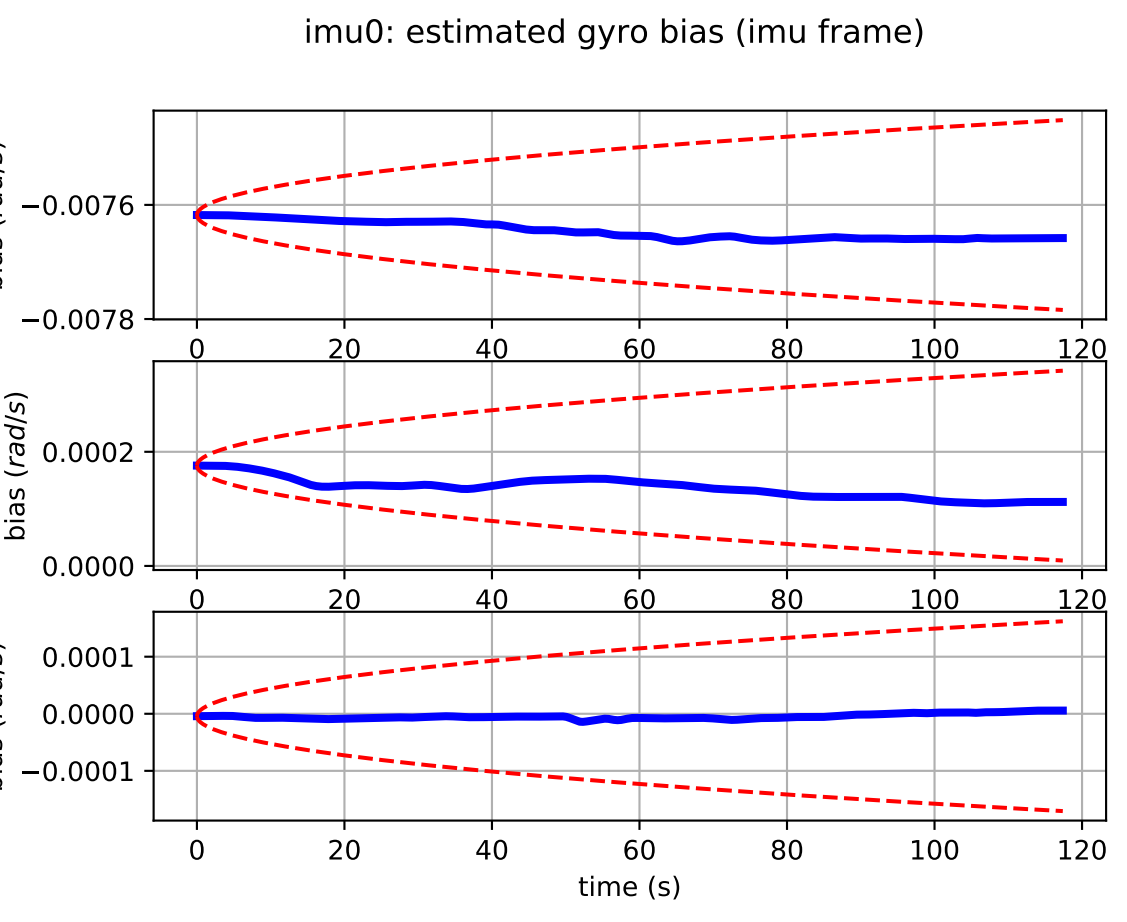

Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
specific force (m/s^2)
specific force (m/s^2)
est. bodyspline
imu0
est. bodyspline
imu0
est. bodyspline
imu0
0
−5
−8
−10
−12
2.5
0.0
−2.5
0
20
40
60
80
100
120
time (s)

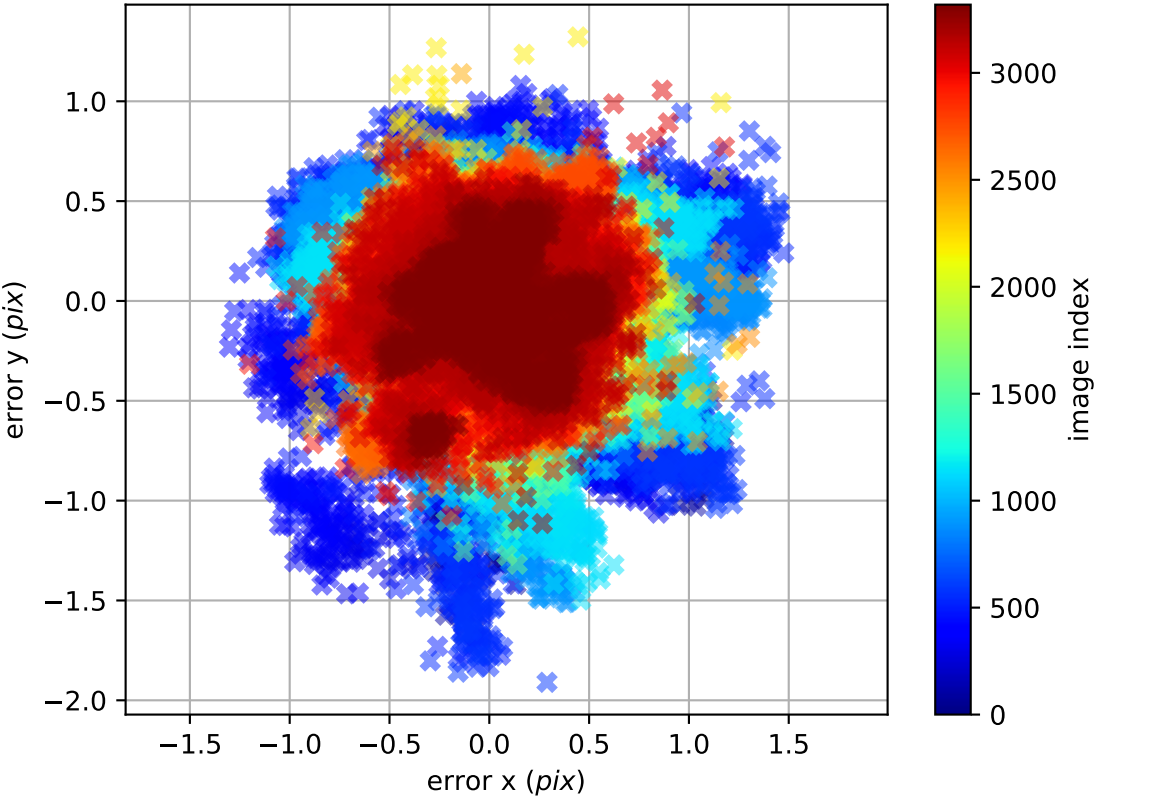

imu0: acceleration error
error (m/s^2)
0.0
−0.5
0
5000
10000
15000
20000
error (m/s^2)
0
−1
0
5000
10000
15000
20000
error (m/s^2)
0.5
0.0
0
5000
10000
15000
20000
error index

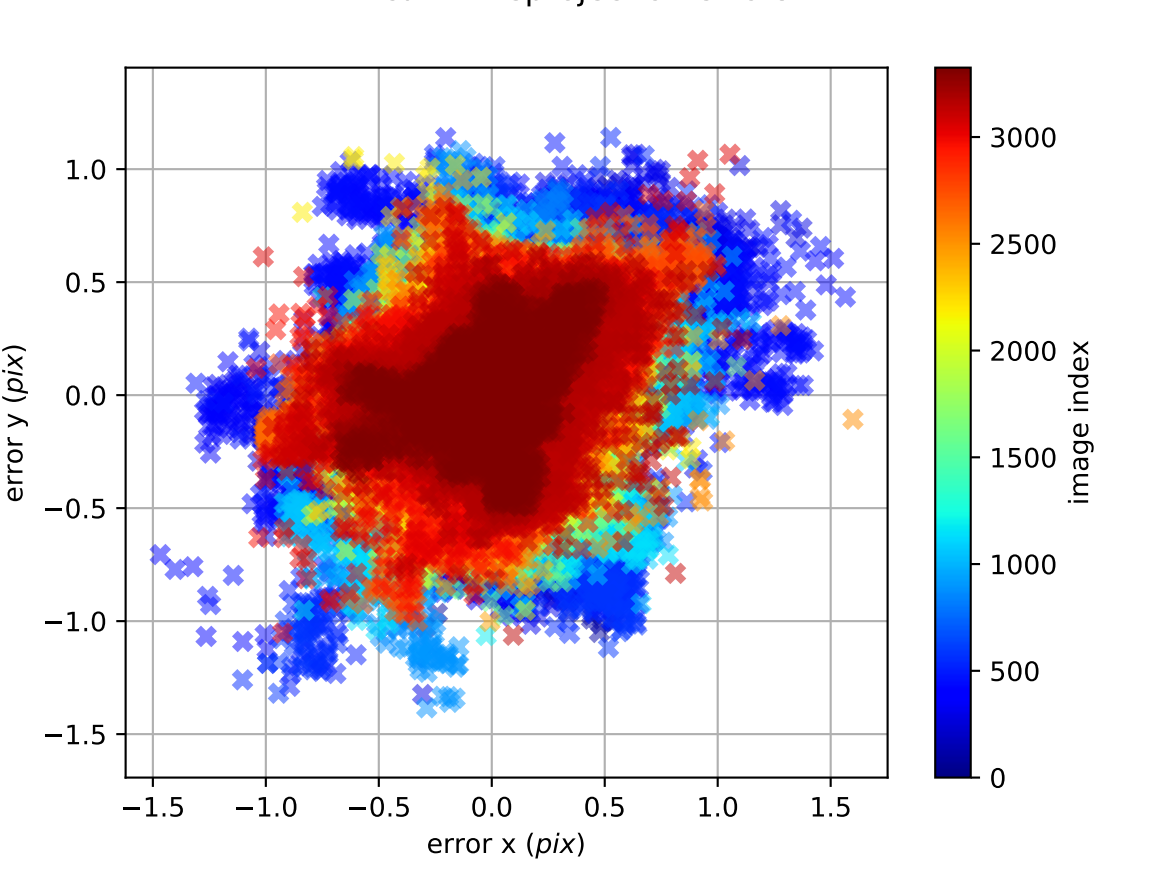

imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
−0.035
−0.040
−0.045
0
20
40
60
80
100
120
bias (m/s^2)
0.065
0.060
−0.1075
−0.1100
−0.1125
time (s)


Comparison of predicted and measured angular velocities (body frame)
ang. velocity (rad/s)
ang. velocity (rad/s)
ang. velocity (rad/s)
est. bodyspline
imu0
est. bodyspline
imu0
est. bodyspline
imu0
1
0
−1
0
20
40
60
80
100
120
time (s)


imu0: angular velocities error
error (rad/s)
0.05
0.00
−0.05
0
5000
10000
15000
20000
error (rad/s)
0.02
0.00
−0.02
0
5000
10000
15000
20000
error (rad/s)
0.025
0.000
−0.025
0
5000
10000
15000
20000
error index


imu0: estimated gyro bias (imu frame)
bias (rad/s)
−0.0076
−0.0078
0
20
40
60
80
100
120
bias (rad/s)
0.0002
0.0000
0
20
40
60
80
100
120
bias (rad/s)
0.0001
0.0000
−0.0001
0
20
40
60
80
100
120
time (s)


cam0: reprojection errors
error x (pix)
error y (pix)
image index
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
3000
2500
2000
1500
1000
500
0


cam1: reprojection errors
error x (pix)
error y (pix)
image index
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
0
500
1000
1500
2000
2500
3000